

JEE MAIN REVISION MASTER: COMMUNICATION SYSTEMS 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 2 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Kcet Communication Systems - Core Application	2 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Communication Systems - Core Application (Standard Yield — 2 PYQs)

Core Principle: Key problem pattern in Kcet Communication Systems.

■■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- **[2018 Shift] Q#1 (MCQ):** "The radius of the Earth is 6400 km . If the height of an antenna is 500 m , then its range is..."
[Ans: N/A]
- **[2017 Shift] Q#2 (MCQ):** "A basic communication system consists of (i) Transmitter (ii) Information sources (iii) User of information (iv) Channel (v) Receiver..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2018 (Q#1)

The radius of the Earth is 6400 km . If the height of an antenna is 500 m , then its range is

- (A) 800 km
- (B) 100 km
- (C) 80 km
- (D) 10 km

Correct Answer: None

Solution Analysis: To find the range of the antenna (i.e., the distance to the horizon), we can use the formula: $d = \sqrt{2Rh}$ where: R is the radius of the Earth h is the height of the antenna Since the Earth's radius is given as 6400 km and the antenna's height is 500 m, we first convert the height into kilometers: $500\text{m} = 0.5\text{km}$ Now, plug the values into the formula: $d = \sqrt{2 \times 6400 \times 0.5} = \sqrt{6400} = 80\text{km}$ Thus, the range of the antenna is 80 km, which corresponds to Option C.